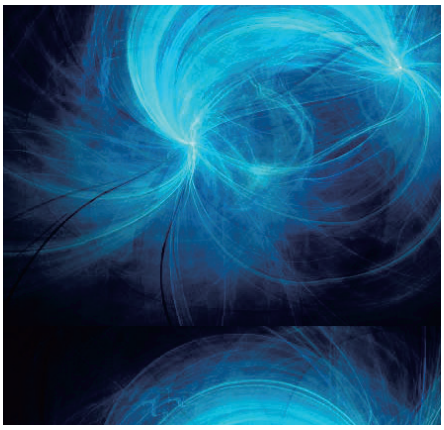

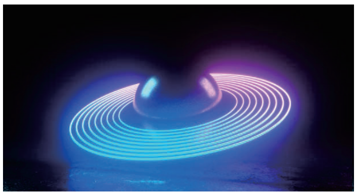

Unit 4: Magnetic Fields

AP Physics C Electricity and Magnetism

TestDaily

Contents

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Force on Moving Charges in Magnetic Fields

TESTDAILY

4.1

Force on Moving Charges in Magnetic Fields

TESTDAILY

4.1 Force on Moving Charges in Magnetic Fields

4.1.1 Magnetic Fields

4.1.1

TESTDAILY

Concept

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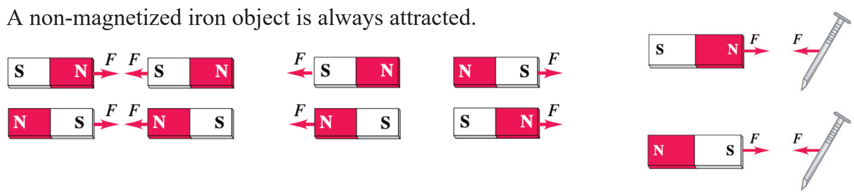
Magnetic Field

- A magnet always has a north pole and a south pole.
- Opposite poles attract; like poles repel.
- A non-magnetized iron object is always attracted.



- A magnet creates a **vector field** around it and another object responds to this field by experiencing a magnetic **force**.

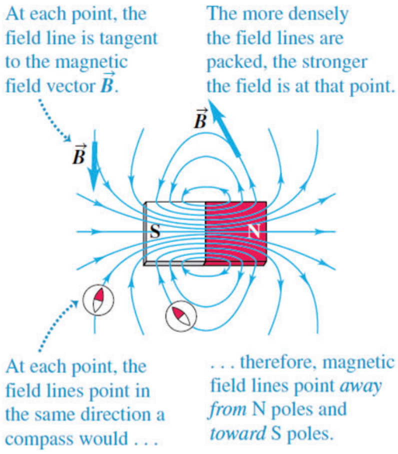
4.1.1

Concept

TESTDAILY

Magnetic Field $\vec{B}$

- Magnetic field lines.
  - Direction
  - Magnitude
- **Always closed loops.**
- A compass (magnetic needle) would point in the direction of magnetic field.

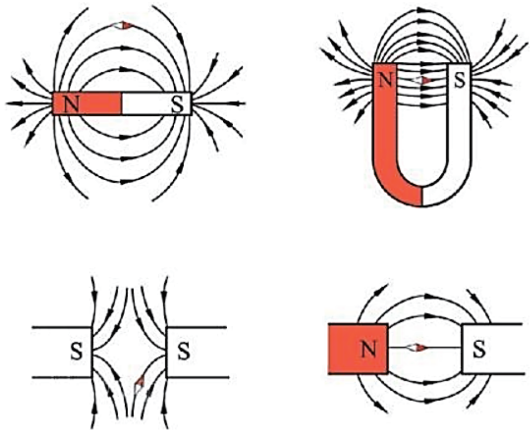


4.1.1

Concept

TESTDAILY

Magnetic Field

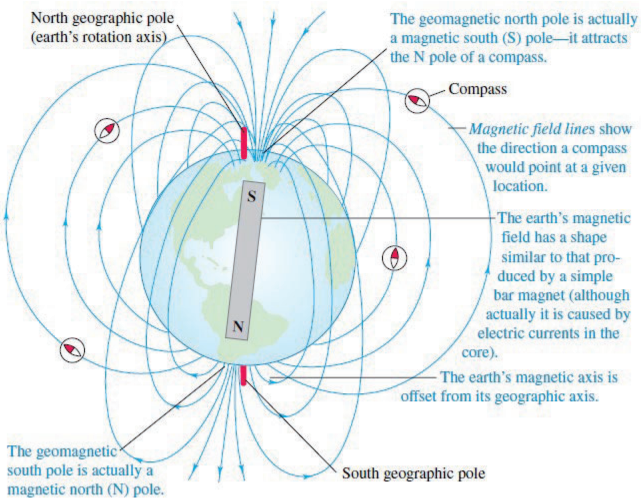


4.1.1

Concept

TESTDAILY

Magnetic Field



4.1.1

Concept

TESTDAILY

Magnetism

- A **moving charge or a current** creates a magnetic **field** in the surrounding space (in addition to its electric field).
- The magnetic field exerts a **force** on any **other moving charge or current** that is present in the field.

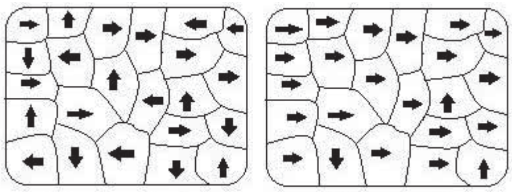
4.1.1

Concept

TESTDAILY

Magnetism

- Electrons orbit around nucleus and they spin, which makes them mini magnets.
- In most materials the magnetism from these mini magnets cancel.
- In iron, nickel, and cobalt, the effects of mini magnets reinforce one another.



Unmagnetized Magnetized

4.1.1

TESTDAILY

4.1 Force on Moving Charges in Magnetic Fields

4.1.2 Magnetic Force on Moving Charges

4.1.2

Concept

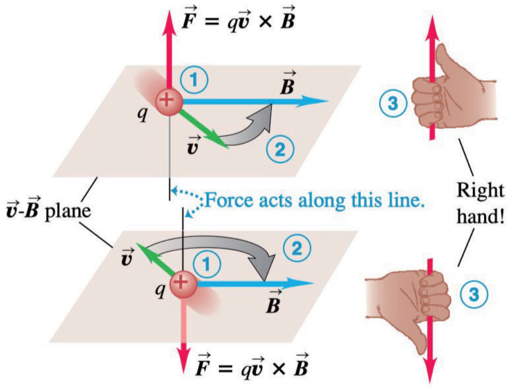
TESTDAILY

Magnetic Force on Moving Charges

- If a particle with charge q moves with velocity $\vec{v}$ through a magnetic field $\vec{B}$, it will experience a magnetic force, $\vec{F}_M$.

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

- Direction of cross-product $\vec{v} \times \vec{B}$: right-hand rule.
    - Fingers curl
    - Thumb points



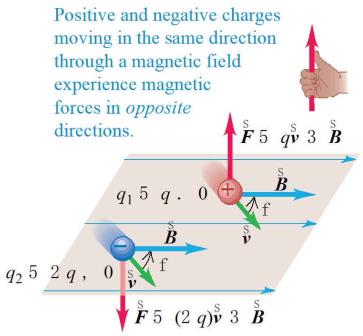
4.1.2

Property

TESTDAILY

Force on Moving Charges

Direction of $\vec{F}_M$ = { same as the direction of $\vec{v} \times \vec{B}$ if q is positive; opposite to the direction of $\vec{v} \times \vec{B}$ if q is negative

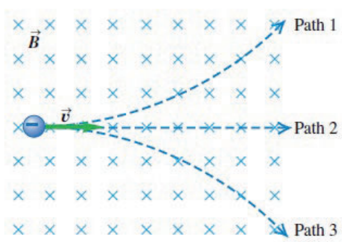


4.1.2

Example

TESTDAILY

The figure at right shows a uniform magnetic field $\vec{B}$ directed into the plane of the paper (shown by the blue ×'s). A particle with a negative charge moves in the plane. Which of the three paths — 1, 2, or 3 — does the particle follow?



4.1.2

Concept

TESTDAILY

Force on Moving Charges

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

- Magnitude: $|\vec{F}_M| = |q|vB\sin\theta = |q|v_{\perp}B$
- Magnetic force is nonzero when the particle is
  - **charged** and
  - **moving** in a direction
  - **nonparallel** to the field
- Lorentz force law

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Physical quantity	Magnetic field 磁场
Definition	A vector that predicts the force on a charged by the Lorentz force law
Symbol	$\vec{B}$
Equation	$\vec{F}_M = q\vec{v} \times \vec{B}$
Unit	tesla, T 1 T=1N/(C·m/s) =1 N/(A·m)

4.1.2

TESTDAILY

4.1 Force on Moving Charges in Magnetic Fields

4.1.3 Uniform Circular Motion in Magnetic Fields

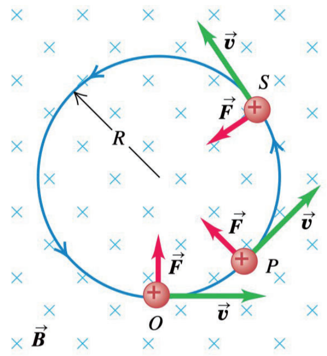
4.1.3

Concept

TESTDAILY

Path of Moving Charges in Uniform Magnetic Fields

- Magnetic force is **always perpendicular to** the velocity, which results in a **uniform circular motion.**

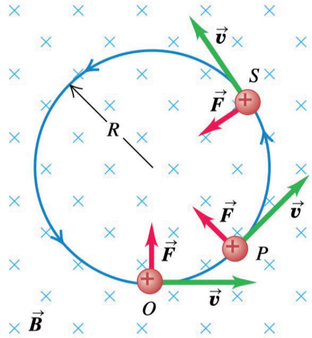


4.1.3

Concept

Path of Moving Charges in Uniform Magnetic Fields

- What is the radius of the circle?
- What is the period of the uniform circular motion?



Exercises

2012_Intl_MCQ_34

The figure above shows the paths of five particles as they pass through the region inside the box that contains a uniform magnetic field ***B*** directed out of the page. Which particle has a positive charge?

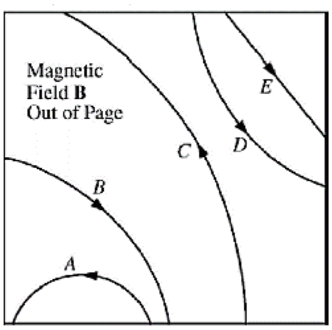
(A) *A*

(B) *B*

(C) *C*

(D) *D*

(E) *E*

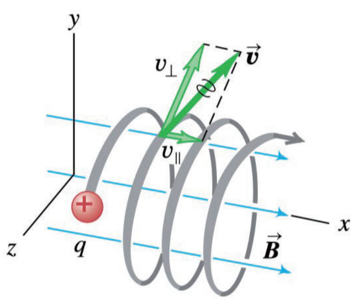


Concept

Path of Moving Charges in Uniform Magnetic Fields

- **Helical** motion

This particle's motion has components both parallel ($v_{\parallel}$) and perpendicular ($v_{\perp}$) to the magnetic field, so it moves in a helical path.



4.1 Force on Moving Charges in Magnetic Fields

4.1.4 Moving Charges in Electric and Magnetic Fields

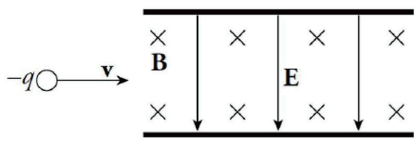
4.1.4

Example

TESTDAILY

Princeton_2020_Chapter15_Example3

A particle of charge $-q$ is shot into a region that contains an electric field, $\boldsymbol{E}$, crossed with a perpendicular magnetic field, $\boldsymbol{B}$. If $E = 2 \times 10^4$ N/C and $B = 0.5$ T, what must be the speed of the particle if it is to cross this region without being deflected?



Answer: 4×10^4 m/s

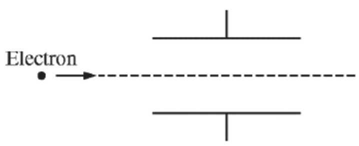
4.1.4

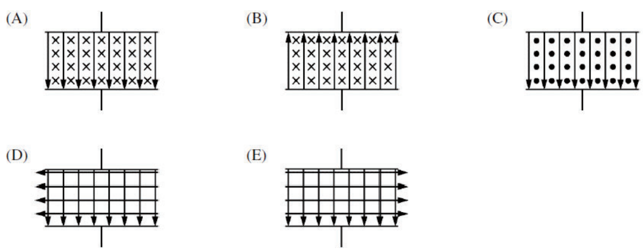
Exercises

TESTDAILY

2016_Intl_MCQ_31

Uniform magnetic and electric fields exist between the two oppositely charged parallel plates shown in the figure above. An electron travels horizontally between the plates. Assuming gravitational effects to be negligible, which of the following diagrams shows a combination of electric and magnetic field directions that will allow the electron to travel undeflected?



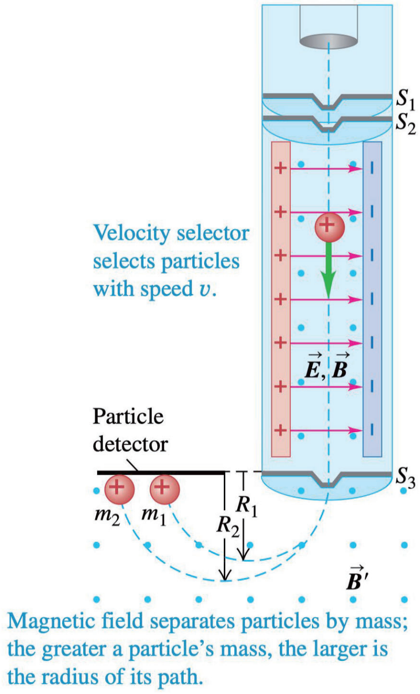


4.1.4

Example

Mass Spectrometer

- Acceleration
- Velocity selection
- Detection



TESTDAILY

4.1.4

Forces on Current-Carrying Wires in Magnetic Fields

TESTDAILY

4.2

Forces on Current-Carrying Wires in Magnetic Fields

TESTDAILY

4.2 Forces on Current-Carrying Wires in Magnetic Fields

4.2.1 Magnetic Force on Current-Carrying Wires

4.2.1

Concept

TESTDAILY

Straight-Line Segment in Uniform Field

- For a single charged particle, the magnetic force is

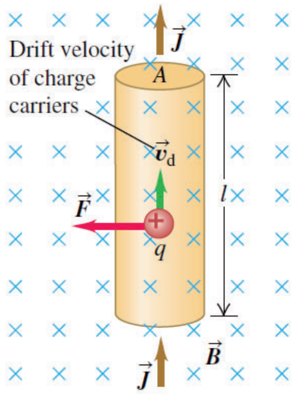
$$\vec{F}_q = q\vec{v}_d \times \vec{B}$$

- The total force on all the moving charges in this segment has magnitude

$$F_{total} = NF_q = (nV)(qv_dB) = (nAl)(qv_dB)$$
$$= (nqv_dA)(lB) = IlB$$

- If the magnetic field is not perpendicular to the wire

$$F = IlB_{\perp} = IlB\sin\phi$$
$$\vec{F}_M = I\vec{l} \times \vec{B}$$

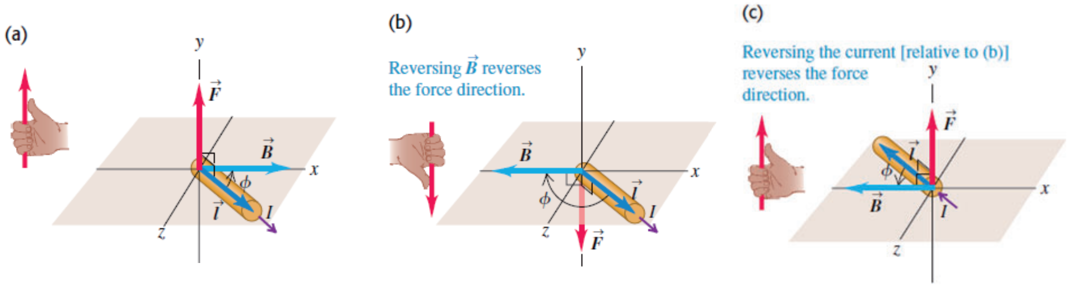


4.2.1

Property

TESTDAILY

Magnetic Force on Current-Carrying Wires



4.2.1

Exercise

TESTDAILY

2016_Intl_MCQ_30

A 0.20 m long wire carries a current of 15 A and lies perpendicular to a magnetic field. The magnetic force on the wire is measured to be 0.060 N. What is the magnitude of the magnetic field?

(A) 0.010 T

(B) 0.020 T

(C) 0.040 T

(D) 0.060 T

(E) 0.080 T

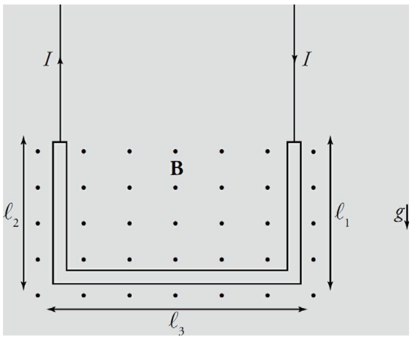
4.2.1

Example

TESTDAILY

Princeton_2020_Chapter15_Example5

A U-shaped wire of mass m is lowered into a magnetic field $\boldsymbol{B}$ that points out of the plane of the page. How much current I must pass through the wire in order to cause the net force on the wire to be zero?



Answer: mg/l_3B

4.2.1

Concept

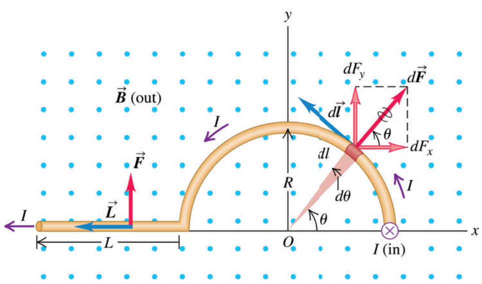
TESTDAILY

Magnetic Forces on Current-Carrying Wires

- If the conductor is not straight,

$$d\vec{F} = Id\vec{l} \times \vec{B}$$

$$\vec{F} = \int Id\vec{l} \times \vec{B}$$

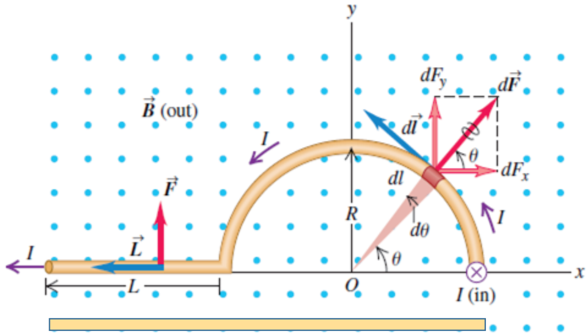


4.2.1

Property

Force on Current-Carrying Wires

- Use **equivalent length** of a straight line connecting both ends.



4.2.1

4.2 Forces on Current-Carrying Wires in Magnetic Fields

4.2.2 Forces and Torques on a Conductive Loop

4.2.2

Concept

Force and Torque on a Current Loop

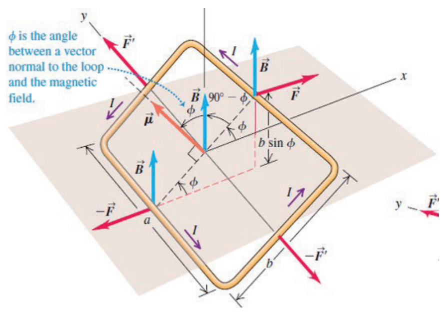
Net force is zero

Net torque is nonzero

$$F = IaB$$

$$\tau_{net} = 2F\frac{b}{2}\sin\phi = (IBa)(b\sin\phi)$$

$$\tau_{net} = IBA\sin\phi$$



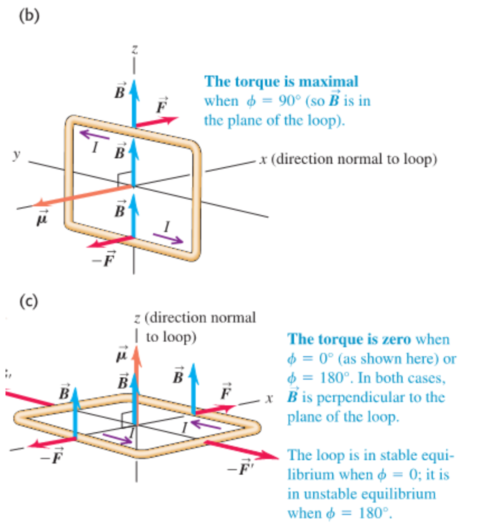
4.2.2

Concept

TESTDAILY

Force and Torque on a Current Loop

Maximum and minimum torque



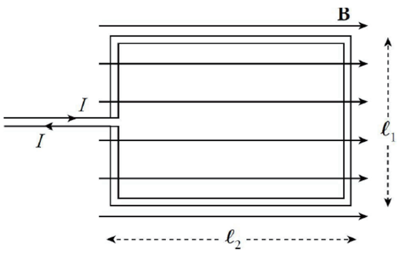
4.2.2

Example

TESTDAILY

Princeton_2020_Chapter15_Example6

A rectangular loop of wire that carries a current I is placed in a uniform magnetic field, $\boldsymbol{B}$, and is free to rotate as shown in the diagram below. What torque does it experience?



Answer: Il_1l_2B

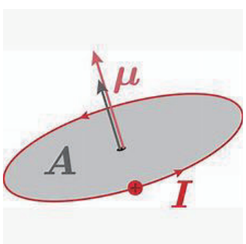
4.2.2

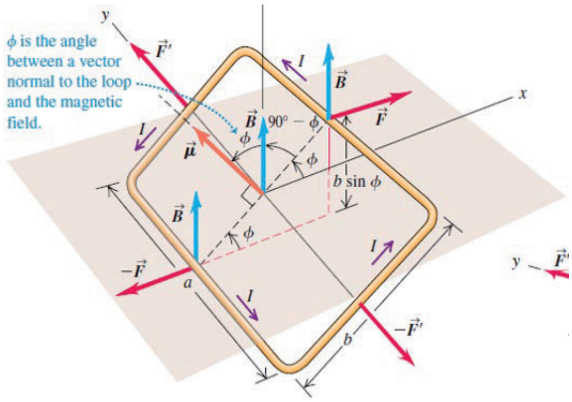
Concept

TESTDAILY

Force and Torque on a Current Loop

Magnetic moment $\vec{\mu} = I\vec{A}$





Vector torque on a current loop

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

4.2.2

Fields of Wires

4.3

Fields of Wires

TESTDAILY

4.3 Fields of Wires

4.3.1 Fields of Long Current-Carrying Wires

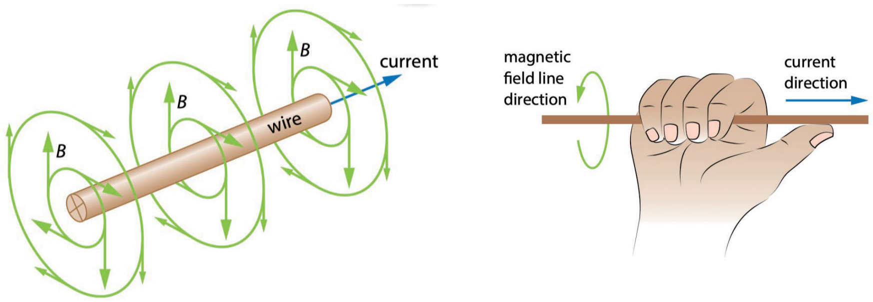
4.3.1

Property

TESTDAILY

Fields of Long Current-Carrying Wires

- Magnetic field lines: circles whose centers are on the wire.
- Direction: a variant of the right-hand rule.



4.3.1

Property

TESTDAILY

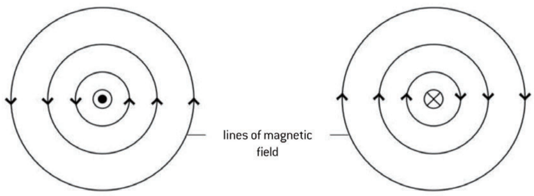
Fields of Long Current-Carrying Wires

- Consider a long, straight wire that carries a current I. The current generates a magnetic field in the surrounding space of magnitude

$$B = \frac{\mu_0}{2\pi}\frac{I}{r}$$

where r is the distance from the wire.

- Vacuum permeability $\mu_0 = 4\pi \times 10^{-7}$ T·m/A



4.3.1

Example

TESTDAILY

Young_Chapter28_Example28.3

A long, straight conductor carries a 1.0 A current. At what distance from the axis of the conductor does the resulting magnetic field have magnitude $B = 0.5 \times 10^{-4}$ T (about that of the earth's magnetic field in Pittsburgh)?

Answer: 4 mm

4.3.1

4.3 Fields of Wires

4.3.2 Magnetic Forces by Current-Carrying Wires

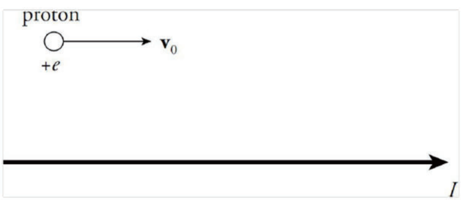
4.3.2

Example

TESTDAILY

Princeton_2020_Chapter15_Example8

The diagram below shows a proton moving with a speed of 2×10^5 m/s, initially parallel to, and 4 cm from, a long, straight wire. If the current in the wire is 20 A, what's the magnetic force on the proton?



Answer: 3.2×10^{-18} N

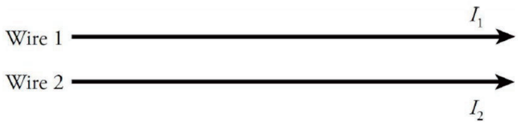
4.3.2

Example

TESTDAILY

Princeton_2020_Chapter15_Example9

The diagram below shows a pair of long, straight, parallel wires, separated by a small distance, d. If currents I_1 and I_2 are established in the wires, what is the magnetic force per unit length they exert on each other?



Answer: $\frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}$

4.3.2

Exercise

TESTDAILY

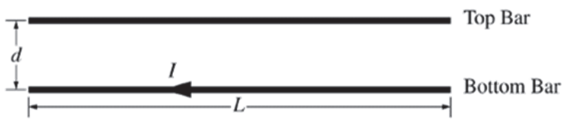
2014_Intl_FRQ_1

Two metal bars of length L are held a vertical distance d apart ($L \gg d$), as shown in the figure above. Each wire carries a current I, and the wires repel each other. The current is to the left in the bottom metal bar.

(a) In the figure above, draw an arrow indicating the direction of the current in the top bar.

(b) In the figure above, use appropriate symbols to indicate the direction of the magnetic field both above and below the bottom bar due to its current I.

(c) Derive an expression for the magnetic force acting on the top bar in terms of I, L, d, and fundamental constants, as appropriate.



4.3.2

4.3 Fields of Wires

TESTDAILY

4.3.3 Superposition of Magnetic Fields

4.3.3

Example

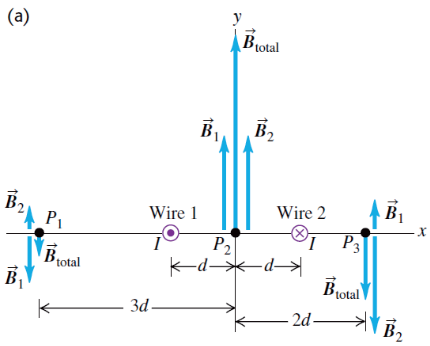
TESTDAILY

Young_Chapter28_Example28.4

Figure on the right is an end-on view of two long, straight, parallel wires perpendicular to the xy-plane, each carrying a current I but in opposite directions.

(a) Find $\vec{B}$ at points P_1, P_2, andP_3.

(b) Find an expression for $\vec{B}$ at any point on the x-axis to the right of wire 2.

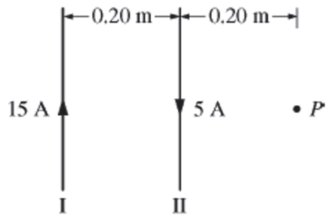


4.3.3

Example

TESTDAILY

2014_Intl_MCQ_24



Two parallel wires that lie in the plane of the page, as shown above, are a distance of 0.20 m apart. Wire I carries a current of 15 A toward the top of the page, and wire II carries a current of 5 A toward the bottom of the page.

What is the magnitude of the net magnetic field at point P, located 0.20 m to the right of wire II?

(A) 0.5 μT

(B) 1.0 μT

(C) 2.5 μT

(D) 5.0 μT

(E) 7.5 μT

4.3.3

Biot-Savart Law and Ampere's Law

TESTDAILY

4.4

Biot-Savart Law and Ampere's Law

TESTDAILY

4.4 Biot-Savart Law and Ampere's Law

TESTDAILY

4.4.1 Biot-Savart Law

4.4.1

Concept

TESTDAILY

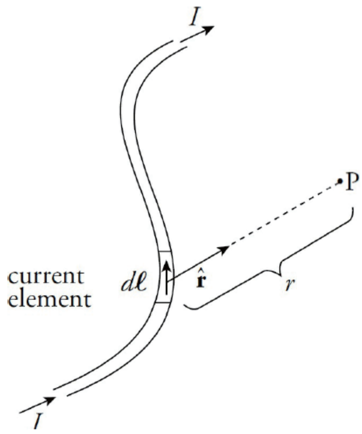
Biot-Savart Law

- Infinitesimal section of current of length dl
- Current element with direction $Id\vec{l}$
- Position vector from current element to point P $\vec{r}$
- Magnetic field at P due to current element
  - Direction

$$d\vec{l} \times \vec{r}$$

  - Magnitude

$$|d\vec{B}| = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$



4.4.1

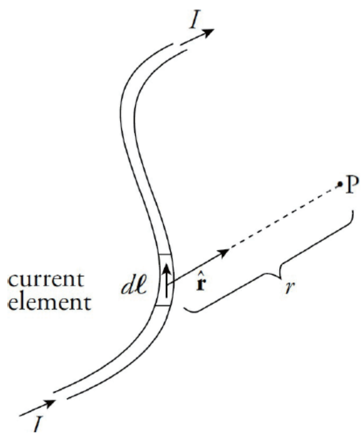
Concept

TESTDAILY

Biot-Savart Law

Unit vector $\hat{r} = \frac{\vec{r}}{r}$

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{l} \times \hat{r})}{r^2}$$



4.4.1

Example

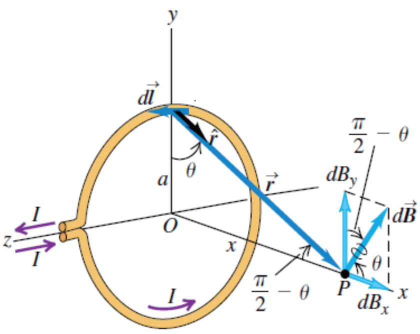
TESTDAILY

Magnetic Field of a Circular Current Loop

- Magnitude:

$$B = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$$

- Direction: right-hand rule.



4.4.1

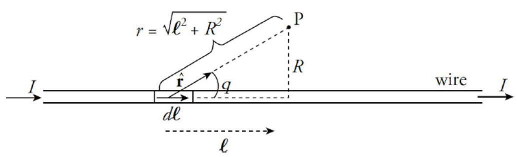
Example

TESTDAILY

Princeton_2020_Chapter15_Example11

Use the Biot-Savart law to show that the magnetic field due to an infinitely long straight wire carrying a current I is given by the equation $B = \frac{\mu_0}{2\pi}\frac{I}{R}$, where R is the distance from the wire. You should use the integration formula:

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{2}{a^2}$$



4.4.1

TESTDAILY

4.4 Biot-Savart Law and Ampere's Law

4.4.2 Ampere's Law

4.4.2

Concept

Ampere's Law

- Analogous to Gauss's Law.

- Line integral of $\vec{B}$ along a path.
  - $\vec{B} \cdot d\vec{l} = Bdl\cos\phi = B_{\parallel}dl$
  - Integrate around **a closed path**, denoted by $\oint \vec{B} \cdot d\vec{l}$.

4.4.2

Concept

TESTDAILY

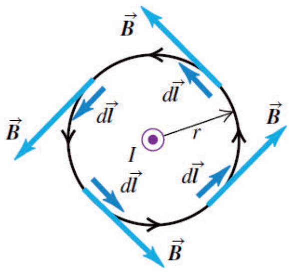
Ampere's Law

Magnetic field parallel to the line segment

$$\oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel}\, dl = B \oint d\, l = \frac{\mu_0 I}{2\pi r}(2\pi r) = \mu_0 I$$

(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



4.4.2

Concept

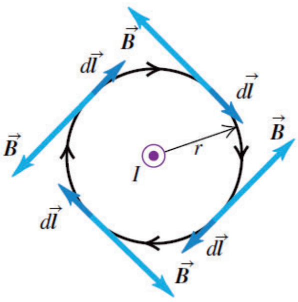
TESTDAILY

Ampere's Law

- Integration path can go around in either direction.
- If magnetic field is in opposite direction, the integral will be negative ($\cos 180° = -1$).

(b) Same integration path as in (a), but integration goes around the circle clockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$

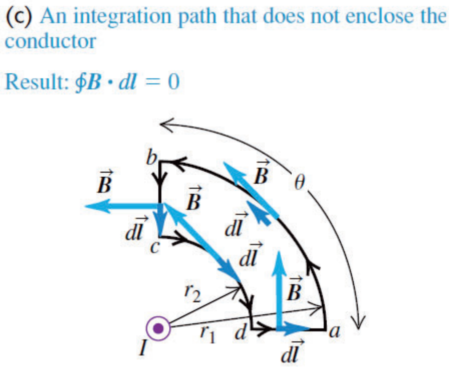


4.4.2

Concept

Ampere's Law

- If the loop does not enclose the current, the integral is zero.



4.4.2

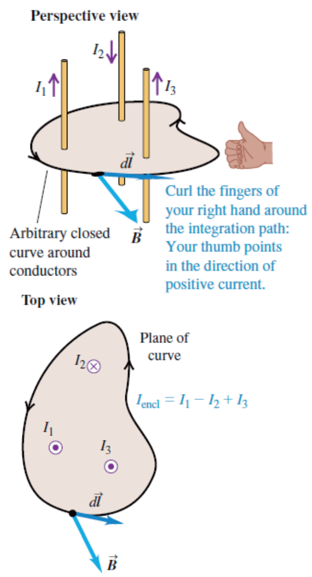
Concept

TESTDAILY

Ampere's Law

- Generalization: several long, straight conductors pass through the surface bounded by the integration path.
- Determine the direction of travel around the Amperian loop, then determine the sign of current by right hand rule.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$



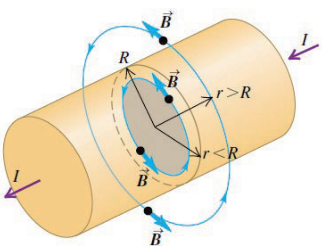
4.4.2

Example

TESTDAILY

Young_Chapter28_Example28.8

A cylindrical conductor with radius R carries a current I. The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnetic field as a function of the distance r from the conductor axis for points both inside ($r < R$) and out-side ($r > R$) the conductor.



Answer: $B = \dfrac{\mu_0 I}{2\pi} \dfrac{r}{R^2}, B = \dfrac{\mu_0 I}{2\pi r}$

4.4.2

Example

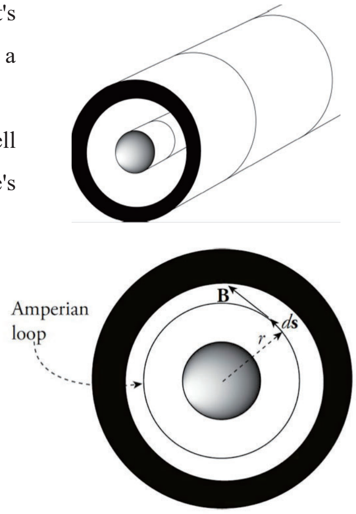
TESTDAILY

Princeton_2020_Chapter15_Example14

The figure below is a cutaway view of a long coaxial cable that's composed of a solid cylindrical conductor (of radius R_1) surrounded by a thin conducting cylindrical shell (of radius R_2).

The inner cylinder carries a current of I_1 and the outer cylindrical shell carries a **smaller** current of I_2 in the opposite direction. Use Ampere's law to find the magnitude of the magnetic field:

(a) in the space between the inner cylinder and the shell



Answer: $B = \frac{\mu_0}{2\pi}\frac{I_1}{r}$

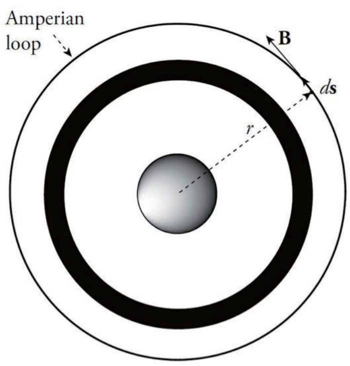
4.4.2

Example

TESTDAILY

Princeton_2020_Chapter15_Example14

(a) outside the outer shell.



Answer: $B = \frac{\mu_0}{2\pi}\frac{I_1 - I_2}{r}$

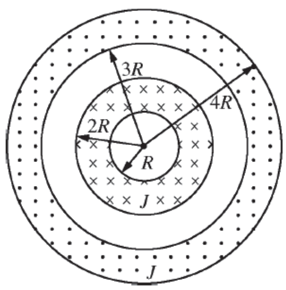
4.4.2

Exercise

TESTDAILY

2016_Intl_MCQ_19

Two long, hollow, concentric conducting cylinders carry currents in opposite directions into and out of the plane of the page, as shown in the cross section above. The currents are unequal, but the current density J is the same for both cylinders. In which of the following regions can the net magnetic field be zero at some nonzero finite distance r from the central axis?



(A) $r < R$ only

(B) Both $r < R$ and $r < R < 2R$

(C) Both $r < R$ and $2R < r < 3R$

(D) Both $r < R$ and $3R < r < 4R$

(E) $r > 4R$ only

4.4.2

4.4 Biot-Savart Law and Ampere's Law

4.4.3 Solenoids

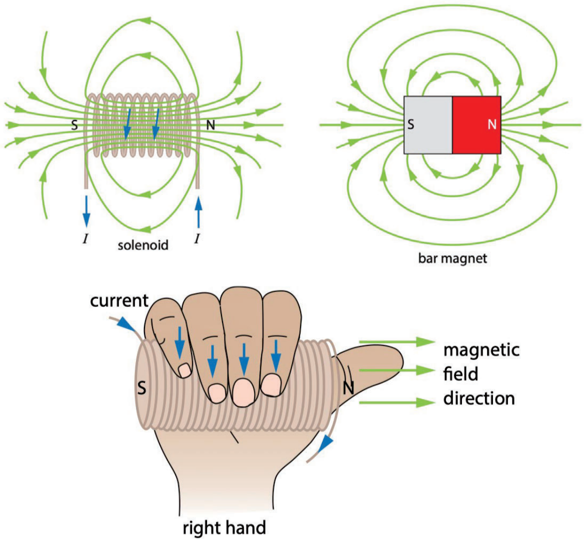
4.4.3

Concept

TESTDAILY

Solenoid

- Tightly wound coils, each considered a circular loop.
- Magnet field similar to a bar magnet.
- Use **right hand rule** to determine the direction of field.

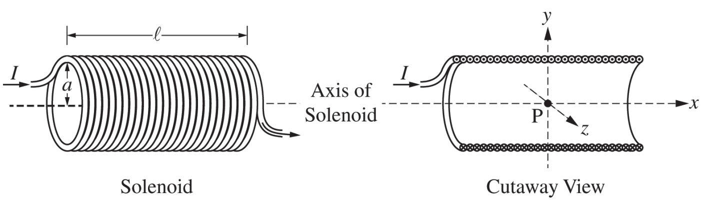


4.4.3

Example

TESTDAILY

2019_NA_Set1_FRQ_3



Note: Figures not drawn to scale.

(a) Select the correct direction of the magnetic field at point P.

______$+x$ -direction ______$+y$ -direction ______$+z$ -direction

______$-x$ -direction ______$-y$ -direction ______$-z$ -direction

Justify your selection.

4.4.3

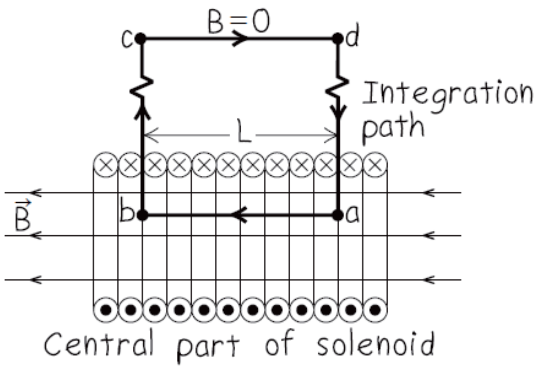
Property

TESTDAILY

Field of a Solenoid

$$B_s = \mu_0 n I$$

- Assumptions:
  - Zero external field
  - Uniform internal field along axis
- Rectangular amperian loop



4.4.3

Example

TESTDAILY

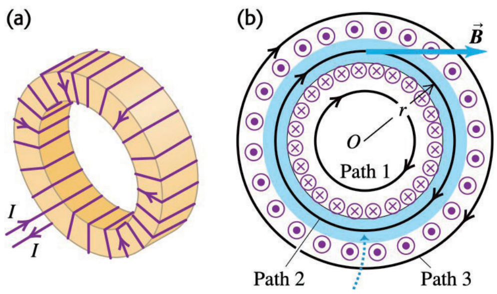
Princeton_2020_Chapter15_Example16

The figure below is a cross section of a toroidal solenoid (a solenoid bent into a doughnut shape) of inner radius R_1 and outer radius R_2. It consists of N windings, and the wire carries a current I. What's the magnetic field in each of the following regions?

(a) $r < R_1$

(b) $R_1 < r < R_2$

(c) $r > R_2$



4.4.3

Answer to MCQ

TESTDAILY

Section	Source	Answer
4.1.3	2012_Intl_MCQ_34	B
4.1.4	2016_Intl_MCQ_31	A
4.2.1	2016_Intl_MCQ_30	B
4.4.2	2016_Intl_MCQ_19	D

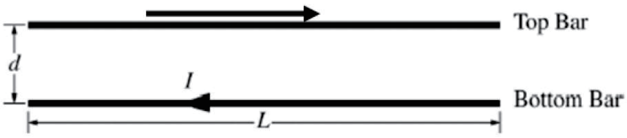
4

Answer to FRQ

2014_Intl_FRQ_1

15 points total **Distribution of points**

(a) 1 point



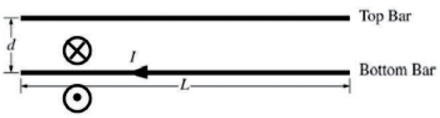
For an arrow indicating that the current must be to the right in the top bar 1 point

Answer to FRQ

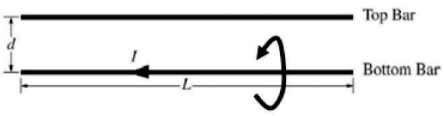
TESTDAILY

2014_Intl_FRQ_1

(b) 2 points



or



For a circle with a dot indicating that the magnetic field is out of the page below the bar as shown in the figure above left 1 point

For a circle with an X indicating that the magnetic field is into the page above the bar as shown in the figure above left 1 point

OR

2 points for an arc indicating the magnetic field is out of the page below the bar and into the page above the bar as shown in the figure above right

Answer to FRQ

TESTDAILY

2014_Intl_FRQ_1

(c) 3 points

For using a correct expression for the magnetic force 1 point

$F = ILB$

For substituting a correct expression for the magnetic field into the above equation 1 point

$F = IL\dfrac{\mu_0 I}{2\pi d}$

For a correct answer 1 point

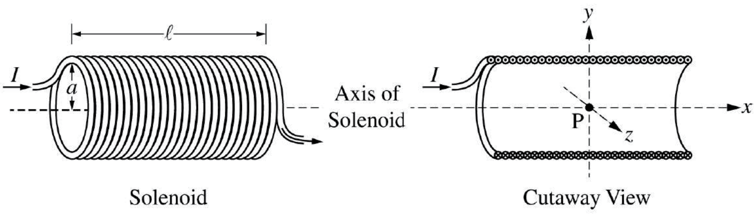
$F = \dfrac{\mu_0 I^2 L}{2\pi d}$

Answer to FRQ

TESTDAILY

2019_NA_Set1_FRQ_3

15 points



Note: Figures not drawn to scale.

A solenoid is used to generate a magnetic field. The solenoid has an inner radius a, length ℓ, and N total turns of wire. A power supply, not shown, is connected to the solenoid and generates current I, as shown in the figure on the left above. The x-axis runs along the axis of the solenoid. Point P is in the middle of the solenoid at the origin of the xyz-coordinate system, as shown in the cutaway view on the right above. Assume $\ell >> a$.

4

Answer to FRQ

TESTDAILY

2019_NA_Set1_FRQ_3

(a) LO CNV-8.E.a, SP 7.A, 7.C
2 points

Select the correct direction of the magnetic field at point P.

____ $+x$-direction ____ $+y$-direction ____ $+z$-direction
____ $-x$-direction ____ $-y$-direction ____ $-z$-direction

Justify your selection.

For choosing the "$+x$-direction" and providing a justification		1 point
For a correct justification		1 point
Example: Using the right-hand rule for current on the left side of the solenoid, the fingers curl into the loop, so the magnetic field points to the right, or in the $+x$-direction.		
Example: Using the right-hand rule for solenoids, when the fingers curl around the solenoid in the direction of the current, the thumb points to the right, therefore the magnetic field is to the right, or in the $+x$-direction.		

4

TESTDAILY

4